SSM Vision Survey and qlfm Backbone Analysis

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Overview

This document summarizes a three-part conversation covering: (1) a survey of state space models (SSMs) adapted for vision tasks; (2) an assessment of whether any of those models could replace the Transformer backbone in the qlfm charge+light foundation model for LArTPC neutrino experiments; and (3) a scaling analysis asking whether Mamba-ND would outscale the Transformer as the model grows.

The context is the `multi-modal-qlmatch` project on MicroBooNE open data, which completed two rounds of experiments. Round 1 pretrained a bidirectional Transformer (MAE objective, charge + PMT light tokens) and showed that pretraining helps decisively at low label budgets and that light is the dominant input for neutrino/cosmic separation. Round 2 compared that backbone against Pandora reconstruction, Panda (PTv3 + Sonata), PoLAr-MAE, FM4NPP (Mamba2), and a self-distillation variant, finding the charge+light MAE Transformer (qlfm-M, 21M) the best on all four physics tasks.

SSM Models Adapted for Vision

The survey covered eight architectures across four clusters. The key design challenge shared by all of them is that SSMs are inherently 1D and causal, so each model must make an explicit choice about how to serialize 2D or 3D spatial data.

Model	Venue	Base SSM	Scan strategy	Architecture	Primary tasks
S4ND	NeurIPS 2022	S4 (LTI)	Factored 2D/3D continuous kernel	Drop-in for Conv2D or self-attention	Image & video classification
Vision Mamba (Vim)	NeurIPS 2024	Mamba (selective)	Bidirectional 1D scan on flattened patches	Flat / isotropic (ViT-style)	Classification, detection, segmentation
VMamba	ECCV 2024	Mamba	4-direction Cross-Scan Module (SS2D)	Hierarchical pyramid (Swin-style)	Classification, detection, segmentation

Model	Venue	Base SSM	Scan strategy	Architecture	Primary tasks
Mamba-ND	ECCV 2024	Mamba	Alternates scan dimension across layers (row-major per dimension)	Flat / isotropic	Images, video, 3D volumes, weather
LocalMamba	AAAI 2025	Mamba	Windowed scan (local then global)	Hierarchical	Classification, detection, segmentation
EfficientVMamba	2024	Mamba	Atrous skip-sampling on VMamba's CSM	Hierarchical, lightweight	Classification, detection, segmentation
PlainMamba	2024	Mamba	Continuous 2D zigzag, direction-aware update	Flat / isotropic	Classification, detection, segmentation
SiMBA	2024	Mamba + EinFFT	Standard scan + Einstein FFT channel mixing	Flat	Vision and time-series
VideoMamba	ECCV 2024	Mamba	Bidirectional B-Mamba on 3D spatiotemporal patches	Flat (ViT-style, 3D)	Video understanding
U-Mamba / SegMamba / VM-UNet	2024	Mamba	Various; U-Net encoder-decoder	Hierarchical, medical	Biomedical image segmentation
FM4NPP	2025	Mamba2 (causal)	Time-major raster, planes interleaved	Flat, per-hit tokens	LArTPC hit regression (k-NNN)

Key finding from the survey. The architectures split along two fault lines. Flat/isotropic models (Vim, Mamba-ND, PlainMamba) are more natural for sequence-style inputs like yours, but lack the spatial inductive bias of hierarchical models. Hierarchical models (VMamba, LocalMamba) assume a dense raster grid, which conflicts with the sparse, irregular wire-time cells in qlfm. FM4NPP is the only model in the survey already operating on LArTPC-style per-hit point clouds with a Mamba backbone.

Can an SSM Replace the qlfm Backbone?

The short answer is: not as a drop-in today, but Mamba-ND is the most architecturally compatible candidate if the pretraining objective is changed. The main obstacle is not speed or model quality — it is the tension between MAE and SSMs.

The MAE–SSM incompatibility. MAE randomly masks an arbitrary visible subset and asks the encoder to contextualize that subset globally. Standard SSMs (causal or bidirectional with a fixed scan) cannot handle this cleanly: removing random tokens changes the positions of all subsequent tokens in the scan, breaking the sequential structure the SSM depends on. Bidirectional SSMs like Vim handle it by scanning all tokens including mask tokens, but this makes the visible-token representations dependent on which mask tokens are present — losing the clean MAE property that only visible tokens participate in the encoder. The practical consequence: **if you keep the MAE objective, the Transformer is the right backbone.**

Assessment of each candidate.

Model	Fit to qlfm input	MAE compatible	Main obstacle
Mamba-ND	Good — alternating-dimension scan handles wire-time grid without assuming a dense raster	Requires fixed visible subset ordering	MAE breaks sequential scan; viable if objective changes to self-distillation or k-NNN
FM4NPP (MambaGPT)	Direct — already operates on LArTPC per-hit tokens; tested in round 2	No — uses k-NNN regression, not MAE	Underperformed at m3 size (4.9M); no multimodal light; causal serialization of cosmic-dominated 2D planes
VMamba / LocalMamba	Poor — assumes dense raster grid; sparse wire-time cells leave gaps in the scan	No	Sparse occupancy incompatible with 4-direction CSM
VideoMamba	Poor — designed for dense 3D video frames	No	Same dense-raster assumption; wrong problem structure
S4ND	Moderate — factored 2D/3D continuous kernel matches wire × time physics	Somewhat — LTI kernel can be used in MAE-style setups	Pre-Mamba (no selectivity); slower per token; no published SSL recipe

Model	Fit to qlfm input	MAE compatible	Main obstacle
SiMBA / PlainMamba	Moderate — flat architecture suits sequence-style input	Partial	Neither designed for sparse point-cloud inputs; no multimodal recipe

What round 2 showed about FM4NPP (the only SSM actually tested). FM4NPP at m3 (4.9M parameters) landed between qlfm-S-C and qlfm-S-CL on all four tasks — useful but not competitive with qlfm-M (21M). The tech note identified three confounders: (1) FM4NPP was ~4× smaller than the other models; (2) causal serialization of cosmic-dominated 2D wire planes is a poor fit for the physics; (3) the k-NNN relative-offset objective is weaker than MAE for event-level features. These are all fixable in a round-3 ablation, but they mean the round-2 result is not a clean SSM-vs-Transformer comparison.

The clearest path to a fair test is an FM4NPP m4 (21M) run with multimodal light tokens interleaved in the scan sequence, compared against qlfm-M at matched compute and parameter count. That is the experiment the round-2 plan already gestures toward.

Scaling Analysis: Mamba-ND vs Transformer

Scaling has two independent axes in this project, and the answer is different on each one.

Axis 1: Token count (cell-level → per-hit tokens). This is where SSMs have a structural, hardware-grounded advantage. Transformer attention scales as $O(N^2)$ in sequence length; Mamba scales as $O(N)$. Moving from qlfm's current ~2k cell tokens to per-hit tokens (~15k/event) is a ~7.5× increase in sequence length. For a Transformer, attention cost grows by ~56×; for Mamba-ND, by ~7.5×. At the per-hit scale, Mamba2 also offers constant-memory inference (no KV cache growth), which is relevant for processing long in-spill event windows. The round-2 FM4NPP run at ~14.5k tokens/event did not hit a memory wall on A100-40GB, confirming the per-hit token scale is feasible with an SSM where it would be expensive with global attention.

Axis 2: Parameter count (6.6M → 21M → larger). This is where the evidence runs against pure SSMs in vision. The round-1 and round-2 experiments showed clean, monotonic scaling with the Transformer: going from S (6.6M) to M (21M) improved all four downstream tasks substantially, with no instability. The vision SSM literature documents a consistent opposite pattern:

Problem	Evidence	Fix required
Training instability (vanishing/exploding gradients) at large sizes	SiMBA, GroupMamba, VMamba all report this	Extra normalization (Mamba-2), EinFFT channel mixing, or distillation
Overfitting as model grows	VideoMamba: pure Mamba overfits when scaled; fixed with self-distillation	Self-distillation from a smaller teacher
Feature map artifacts (high-norm background tokens)	Vision Mamba (Vim), Mamba-R: artifacts prevalent even at tiny size, worsen at scale	Register tokens (Mamba-R) or CAM operator (GroupMamba)
Hard ceiling on pure-SSM scale	Most vision Mamba papers stop at ~80–100M parameters without special techniques	Distillation-free methods (StableMamba) needed to reach 300M+

The Transformer in qlfm showed none of these problems between S and M. This matters concretely: the round-1 tech note found that capacity and compute are the primary lever at this scale (not more data), which means parameter-count scaling is exactly the axis you will want to push in round 3.

The language-modeling caveat. In language modeling, Mamba does match Transformer scaling laws on perplexity from 125M to 1.3B parameters. This result does not transfer to vision because vision SSMs process non-sequential 2D/3D data via scan heuristics, which introduces instabilities not present in causal left-to-right text. The best-performing large SSM systems in both language and vision are now **hybrids**: NVIDIA's Mamba-2-Hybrid (43% SSM + 7% attention + 50% MLP) outperforms a matched pure Transformer at 8B scale; MambaVision (CVPR 2025) uses Mamba in early stages and Transformer attention in later stages.

Conclusions and Recommended Path Forward

Three conclusions follow directly from the evidence.

1. Keep the Transformer for parameter-count scaling. qlfm-M (21M) is the best model on all four tasks in round 2. The Transformer scaled cleanly from 6.6M to 21M with no instability and consistent downstream improvements. The vision SSM literature shows

pure SSMs require substantial extra engineering (distillation, register tokens, special normalization) to reach the same parameter range, with no demonstrated quality advantage once those fixes are applied. The case for replacing the Transformer on this axis is weak.

2. Introduce Mamba-ND (or Mamba2) at the per-hit token scale. When the token count grows from ~2k cell tokens to ~15k per-hit tokens, the $O(N^2)$ attention cost becomes the primary constraint. That is the natural point to test an SSM backbone — not as a full replacement today, but as a targeted ablation at matched parameter count and compute budget. Mamba-ND's alternating-dimension scan is the most architecturally compatible design for the wire × time grid without requiring a dense raster.

3. Change the objective if moving to an SSM backbone. The MAE objective is incompatible with SSMs when random masking is applied, because random removal of tokens disrupts the sequential scan structure. A round-3 SSM ablation should use either k-NNN regression (FM4NPP-style) or self-distillation (Panda-style) as the pretraining objective. The round-2 finding that qlfm-SD underperformed was attributed to an inherited learning rate (~3× too low), not to the objective itself — a corrected qlfm-SD is a viable comparison point.

Suggested round-3 experiments, in priority order.

Priority	Experiment	What it answers
1	FM4NPP m4 (21M) with multimodal light tokens, matched compute	Fair SSM vs Transformer comparison at per-hit scale, same parameter count
2	qlfm-M → larger Transformer (50–100M), same MAE objective	Whether the Transformer continues to scale cleanly beyond 21M
3	qlfm-SD re-run with corrected learning rate (~3× higher)	Whether self-distillation on qlfm cell tokens competes with MAE at same compute
4	Mamba-ND backbone with self-distillation objective, 21M, per-hit tokens	Direct test of Mamba-ND's per-hit-scale advantage with an SSM-compatible objective

The two scaling axes point toward a hybrid architecture as the likely long-run optimum: a Transformer for parameter-count scaling and global cross-modal attention (charge ↔ light), with SSM layers handling the long within-event per-hit sequence. That is consistent

with what the broader literature has converged on (NVIDIA Mamba-2-Hybrid, IBM Granite 4.0, MambaVision).